

ANUBHAV AGRAWAL

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TECHNICAL SKILLS

Languages & Tools: Python, Bash, Git, Linux

Model Optimization: PyTorch, ONNX (dynamic shapes, control flow, opsets), ONNX Runtime, TensorRT, trtexec

Quantization & On-Device: INT8/FP16 PTQ with calibration, 4-bit GGUF/MLX (llama.cpp, mlx-lm)

ML Frameworks: PyTorch, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas

Platform: Docker, GitLab CI/CD, AWS (EC2), FastAPI, Grafana/Loki/Mimir/Tempo, MongoDB, SQLite

EXPERIENCE

Griphic

Delhi

AI Engineer II

Jan 2026 – Present

- Sole engineer on a multi-agent LLM service in production (Python/FastAPI) that turns interviews into structured blueprints, end to end.
- Cut per-request generation cost by **60%** by tracking what every token costs, caching prompts, and bounding parallel fan-out.
- Made outputs safe to ship unread: schema-validated structured outputs with bounded self-correction (**99.5%** ship with zero human review), plus a two-stage guardrail — batched judge, then independent adjudicator — that stops prompt injection and credential leaks for at most **2 extra LLM calls** per clean request.
- Quality gates as code: every merge clears unit + BDD suites, SonarQube ($\geq 90\%$ coverage, VAPT/ISO 27001-aligned), and Trivy HIGH/CRITICAL scans.

Innvolution Healthcare Pvt. Ltd.

Bangalore

Machine Learning Engineer & Team Lead

Sep 2023 – Dec 2025

- Owned AIMAG, an X-ray super-resolution and denoising product at **\$200K+ ARR**, from scratch to serving — datasets, model R&D, quantization, and release after clinical trials.
- Made the model real-time: PyTorch $\rightarrow$ ONNX $\rightarrow$ TensorRT with INT8/FP16 **post-training quantization**, debugging dynamic shapes, unsupported ops, and opset mismatches in export; **sub-5 ms per frame** for 4 $\times$ upsampling on an RTX 4090, quantized accuracy checked against the FP32 baseline.
- Built the QCA pipeline (segmentation $\rightarrow$ calibration $\rightarrow$ prediction) on a fine-tuned U-Net++ (ResNet50 + attention); live cardiac-catheterization analysis completes in under 4 seconds.
- Co-inventor on a **granted Indian patent** (No. 604176, AI-Powered X-Ray Image Enhancement) and **2 pending applications**: blind single-image super-resolution, DSA roadmap generation, and real-time fluoroscopy enhancement.
- Led **6 ML engineers**: owned the technical roadmap, set up architecture and code review, and wrote each release's deployment runbook (calibration steps, accuracy gates, latency budgets).

PROJECTS

ipic — On-Device Multimodal Search Engine

[GitHub](#) | [Write-ups](#)

Fully offline search across text, PDFs, images, audio, and video — no cloud, no telemetry

2026

- Architected a sharded index — one self-contained SQLite + vector-store shard per root — under a supervising engine that parallelizes scan, embedding, and search; BGE and CLIP run entirely on-device.
- Fused dense (bge-small), CLIP vision, BM25 (FTS5), and acoustic lanes via reciprocal rank fusion over custom mmap'd i8-quantized vector stores; exact scan beats HNSW at file-index scale: **26–32 ms** queries, sub-millisecond vector scans.
- Scaled to a **350k-file** home directory at **$\sim 2.5\text{k files/s}$** with bounded-queue backpressure, batched shard claims, and a per-stage compute budget — eliminating the SQLite convoy and queue growth that capped v1.

Inference Lab — LLM Serving Benchmarks on Apple Silicon

[GitHub](#) | [Write-up](#)

Reproducible benchmark harness: mlx-lm vs llama.cpp under a single fairness contract

2026

- Built a reproducible benchmark harness holding mlx-lm and llama.cpp to one fairness contract (fixed prompts, greedy decoding, matched budgets): **45–56 tok/s** single-stream for Qwen2.5-7B-Instruct (4-bit) on an M4 Pro.
- Profiled both engines and isolated why throughput stays **flat from 1 to 32 concurrent users**: a per-instance lock serializes requests inside each serving engine.
- Proved decode is memory-bandwidth-bound: K process-isolated instances behind one dispatcher return only **+9%** for **3 $\times$ the memory** — a single instance already saturates the M4 Pro's ~ 273 GB/s bus.

OPEN SOURCE

ONNX Runtime — merged a ReshapeFusion graph-optimizer crash fix ([#33091](#)); open PRs in **PyTorch** (ONNX int32-quantization export, [#200180](#)) and **mlx-lm** (prompt-cache byte budget, [#1944](#)).

EDUCATION

BITS Pilani — M.Tech in Artificial Intelligence & Machine Learning, WILP

2025 – 2027 (expected)

Bhilai Institute of Technology, Durg — B.Tech in Computer Science Engineering

2019 – 2023